

Brief Summary: One-Hot Encoding vs. Label Encoding (Categorical Encoding)

Categorical Encoding refers to the process of converting non-numeric categorical variables (e.g., color, education) into numeric formats compatible with machine learning algorithms. Two core methods are One-Hot Encoding and Label Encoding, differing in logic, pros/cons, and use cases:

1. Label Encoding

- **Basic Concept:** Maps each unique category to an integer (e.g., "Low"→0, "Medium"→1, "High"→2).
- **Pros:** Simple to implement; preserves dimensionality (no feature explosion); retains ordinal relationships.
- **Cons:** Introduces false numerical order for nominal variables (e.g., "Red"→0, "Blue"→1 implies "Red < Blue", which is meaningless).
- **Use Cases:** Ordinal categorical variables (with inherent rank/order), such as education level (High School→0, Bachelor→1, Master→2), customer satisfaction scores (1→0, 2→1, ..., 5→4), or product quality grades (Poor→0, Fair→1, Good→2).

2. One-Hot Encoding

- **Basic Concept:** Creates a binary (0/1) feature for each unique category; a "1" indicates the presence of the category (e.g., "Red"→[1,0,0], "Blue"→[0,1,0], "Green"→[0,0,1]).
- **Pros:** Eliminates false order for nominal variables; directly interpretable.
- **Cons:** Causes "curse of dimensionality" for high-cardinality variables (e.g., zip codes with 1000+ categories); increases computational complexity.
- **Use Cases:** Nominal categorical variables (no inherent order), such as color (Red/Blue/Green), city (Beijing/Shanghai/Guangzhou), gender (Male/Female/Other), or product type (Laptop/Phone/Tablet).

Key Distinction

Choose Label Encoding for **ordinal variables** (to preserve order) and One-Hot Encoding for **nominal variables** (to avoid false numerical relationships).